

- 4 (a) Alternating voltage can be stepped up or down whereas direct voltage cannot be stepped up or down easily. Transmission of power at high voltages minimizes power loss. B2
- (b) Root-mean-square value of an alternating current is defined as the value of the steady direct current which would dissipate heat at the same rate in a given resistance as the alternating current. B2

Comments:

‘Square root of the mean of the square of the instantaneous a.c. current’ is not acceptable as explanation for r.m.s. value.

- (c) Step-up transformer with turns ratio $\frac{N_s}{N_p} = \frac{240}{120} = 2.0$ B1

Comments:

The transformer is required for stepping up the 120 V provided in USA to the 240 V required by the hair dryer (connected to the secondary coils).

- (d) Assuming no power losses in the transformer, B1
 $120I_p = 1000$ M1
 $I_p = 8.33$ A r.m.s.
Peak current = $8.33\sqrt{2} = 11.8$ A A1

Comments:

Do not accept: resistance remains constant, ideal transformer, a.c. vary sinusoidally